

nccl-tests 1-node collective summary

- Generated: 2026-07-14 11:05:08
- Nodes: node5500, node5502 (each 8 x NVIDIA B200, single node, intra-node NVLink)
- Config: 1 thread, 1 MiB-16 GiB, 5 warmup + 20 iters
- Collectives: sendrecv, reduce, broadcast, gather, scatter, reduce_scatter, all_gather, all_reduce, alltoall, hypercube
- Reference: MIT aicr-benchmarks results_b200.md, Table 1 (b0027, 8x B200, NVLink 5.0 / NVSwitch), busbw at 900 GB/s NVLink max

Converged busbw = busbw at the largest message size, best of out-of-place / in-place (matches the reference methodology). busbw (bus bandwidth) is the figure of merit.

Converged bus bandwidth by collective (GB/s)

Collective	node5500	node5502	Reference (b0027)	node5500 % of ref	node5502 % of ref	Correctness
sendrecv	665.6	664.8	666	100%	100%	PASS
reduce	687.6	685.6	701	98%	98%	PASS
broadcast	675.3	680.0	691	98%	98%	PASS
gather	717.7	716.9	717	100%	100%	PASS
scatter	734.4	734.7	746	98%	98%	PASS
reduce_scatter	694.9	694.1	695	100%	100%	PASS
all_gather	677.9	677.3	684	99%	99%	PASS
all_reduce	838.5	837.9	841	100%	100%	PASS
alltoall	661.4	659.5	675	98%	98%	PASS
hypercube	FAILED	FAILED	—	—	—	FAIL

Bus bandwidth vs message size (out-of-place busbw, GB/s)

sendrecv

Message size	node5500	node5502
1 MiB	28.0	23.8
4 MiB	62.5	63.4
16 MiB	78.5	77.6

64 MiB	84.8	84.6
256 MiB	332.6	332.6
1 GiB	644.4	644.2
4 GiB	661.1	660.7
16 GiB	665.6	664.8

reduce

Message size	node5500	node5502
1 MiB	25.1	19.5
4 MiB	92.5	79.4
16 MiB	298.4	278.5
64 MiB	501.8	507.3
256 MiB	619.8	618.7
1 GiB	660.9	663.6
4 GiB	675.4	677.8
16 GiB	687.6	685.6

broadcast

Message size	node5500	node5502
1 MiB	24.7	18.8
4 MiB	84.2	70.6
16 MiB	303.9	251.2
64 MiB	495.1	493.9
256 MiB	607.1	609.6
1 GiB	642.5	644.4
4 GiB	658.1	663.8
16 GiB	675.3	679.9

gather

Message size	node5500	node5502
1 MiB	23.6	18.6
4 MiB	87.3	81.4
16 MiB	340.7	320.9
64 MiB	613.5	609.7
256 MiB	683.4	684.3
1 GiB	700.4	697.2
4 GiB	715.1	715.9
16 GiB	717.7	716.9

scatter

Message size	node5500	node5502
1 MiB	24.3	19.4
4 MiB	99.1	80.2
16 MiB	378.9	315.5
64 MiB	601.0	576.8
256 MiB	689.4	637.7
1 GiB	726.3	715.0
4 GiB	730.2	731.0
16 GiB	734.4	734.7

reduce_scatter

Message size	node5500	node5502
1 MiB	18.8	14.9
4 MiB	78.1	60.6
16 MiB	145.5	143.8
64 MiB	414.0	415.5
256 MiB	587.5	587.8
1 GiB	642.8	636.1
4 GiB	679.8	680.4

16 GiB	694.8	693.7
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all_gather

Message size	node5500	node5502
1 MiB	16.5	12.4
4 MiB	70.0	53.1
16 MiB	137.2	137.7
64 MiB	416.0	414.9
256 MiB	580.4	582.7
1 GiB	616.7	617.9
4 GiB	652.9	653.2
16 GiB	669.0	669.5

all_reduce

Message size	node5500	node5502
1 MiB	34.5	27.8
4 MiB	126.2	124.3
16 MiB	267.6	265.9
64 MiB	421.0	419.9
256 MiB	655.0	656.1
1 GiB	732.7	732.5
4 GiB	827.2	826.8
16 GiB	838.5	836.5

alltoall

Message size	node5500	node5502
1 MiB	15.2	11.6
4 MiB	57.4	45.5
16 MiB	222.6	178.7

64 MiB	416.6	419.1
256 MiB	532.3	531.8
1 GiB	604.9	602.8
4 GiB	647.4	645.8
16 GiB	660.8	659.0

hypercube

No data (run failed or produced no rows).